

A Complete Bound Action–Frequency and Closed-Orbit Atlas for the Hénon Isochrone Potential

Route-A source-local certificate HCS-C295

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Abstract

We close the bound action–frequency geometry of the planar Hénon isochrone Hamiltonian, with its exact circular-energy boundary. An independent apsidal integral yields the frequency map and the exact noncircular closure criterion; circular, center-crossing, escape, signed angular-momentum, and Kepler-limit faces are separated explicitly. A 116-cell exact certificate, independent symbolic and quadrature checks, 87/87 hostile rejections, and a conservative Route-A audit accompany the proof. The finite grid is not the proof.

1 Model, conventions, and provenance

Let $\mu, b > 0$. In polar variables the unit-mass Hamiltonian is

$$H(r, p_r, L) = \frac{p_r^2}{2} + \frac{L^2}{2r^2} - \frac{\mu}{b + \sqrt{b^2 + r^2}}. \quad (1)$$

Put $\ell = |L|$,

$$A = \sqrt{\ell^2 + 4\mu b}, \quad B = \ell + A, \quad (2)$$

and use physical Hamiltonian time. For $\ell > 0$, the radial action is

$$J_r = \frac{1}{\pi} \int_{r_p}^{r_a} \sqrt{2(E - U_\ell(r))} \, dr, \quad U_\ell(r) = \frac{\ell^2}{2r^2} - \frac{\mu}{b + \sqrt{b^2 + r^2}}. \quad (3)$$

At $\ell = 0$ we use its continuous half-line limit and discuss the full Cartesian return separately.

The potential and its orbit mechanics are classical, originating in Hénon’s two 1959 papers [1, 2]. Ramond and Perez give a modern isochrone-mechanics treatment [3]; Fouvry and Prunet record the frequency map in their official Appendix C [4]. We rederive the formulas to close a reproducible certificate and make no claim of literature priority.

2 Complete bound action theorem

Theorem 1 (Energy, action, and radial period). *For fixed $\mu, b > 0$ and $\ell \geq 0$, define*

$$s_c = \frac{B^2}{4\mu}, \quad r_c^2 = s_c^2 - b^2, \quad E_c = -\frac{\mu}{2s_c} = -\frac{2\mu^2}{B^2}. \quad (4)$$

Bound motion exists exactly for $E_c \leq E < 0$. On this domain,

$$J_r = \frac{\mu}{\sqrt{-2E}} - \frac{B}{2}, \quad (5)$$

$$H(J_r, \ell) = -\frac{\mu^2}{2(J_r + B/2)^2}, \quad (6)$$

$$T_r = \frac{2\pi\mu}{(-2E)^{3/2}}, \quad \Omega_r = \frac{(-2E)^{3/2}}{\mu}. \quad (7)$$

In particular, $J_r \geq 0$ is equivalent to $E \geq E_c$ inside $E < 0$, and T_r is independent of angular momentum.

Proof. Write $s = \sqrt{b^2 + r^2} \geq b$. For $\ell > 0$, differentiation of

$$U_\ell(s) = -\frac{\mu}{b+s} + \frac{\ell^2}{2(s^2 - b^2)}$$

gives the circular equation $\mu(s-b)^2 = \ell^2 s$. Its unique solution above b is $s_c = B^2/(4\mu)$, and substitution gives (4). The effective potential tends to $+\infty$ at zero, to 0^- at infinity, and has only this critical point, hence this is its global minimum. When $\ell = 0$, the potential increases from $-\mu/(2b)$ to zero, and the same formulas give the central minimum. This proves the asserted energy domain.

For the period, set

$$x = b + \sqrt{b^2 + r^2}, \quad r^2 = x(x-2b), \quad Q(x) = 2Ex^2 + (2\mu - 4bE)x - (4\mu b + \ell^2). \quad (8)$$

A direct substitution in the energy equation gives

$$dt = \frac{x-b}{\sqrt{Q(x)}} dx. \quad (9)$$

For a noncircular bound orbit, put $q = -2E > 0$ and write $Q = q(x_a - x)(x - x_p)$. Vieta's identities give

$$x_p + x_a = 2b + \frac{2\mu}{q}, \quad x_p x_a = \frac{A^2}{q}. \quad (10)$$

The arcsine substitution on $[x_p, x_a]$ therefore yields

$$\int_{x_p}^{x_a} \frac{x-b}{\sqrt{Q(x)}} dx = \frac{\pi}{\sqrt{q}} \left(\frac{x_p + x_a}{2} - b \right) = \frac{\pi\mu}{q^{3/2}}.$$

Doubling proves (7); the circular value follows by continuity.

Finally, differentiation of the one-dimensional action integral gives $\partial_E J_r = T_r/(2\pi) = \mu/(-2E)^{3/2}$. Thus $J_r = \mu/\sqrt{-2E} + C(\ell)$. The circular cycle collapses, so $J_r(E_c, \ell) = 0$; since $\mu/\sqrt{-2E_c} = B/2$, this fixes $C = -B/2$ and proves (5). Inversion gives (6), and differentiating it in J_r gives Ω_r . $\square$

Below the circular minimum no real radial interval exists. At $E = 0$ the motion reaches the marginal escape threshold, while $E > 0$ is unbound. Equations (5) and (7) show that both action and period diverge as $E \uparrow 0$; the threshold is not a finite action point.

3 Apsidal map and noncircular closure

Theorem 2 (Frequency map and closure). *For $L \geq 0$ and $\ell > 0$,*

$$\frac{\Omega_\phi}{\Omega_r} = \beta(\ell) = \frac{1}{2} \left(1 + \frac{\ell}{\sqrt{\ell^2 + 4\mu b}} \right), \quad \frac{1}{2} < \beta < 1. \quad (11)$$

The azimuthal advance in one radial period is $2\pi\beta$. A noncircular bound phase-space orbit closes if and only if $\beta \in \mathbb{Q}$. If $\beta = p/q$ in lowest terms, its primitive period is qT_r .

Proof. Because $\dot{\phi} = \ell/r^2$, equations (9) and (8) reduce the half-cycle angular integral by

$$\frac{x-b}{x(x-2b)} = \frac{1}{2} \left(\frac{1}{x} + \frac{1}{x-2b} \right). \quad (12)$$

Besides (10), Vieta gives the shifted product

$$(x_p - 2b)(x_a - 2b) = \frac{\ell^2}{q}. \quad (13)$$

Using $\int_{x_p}^{x_a} dx/[x\sqrt{(x_a-x)(x-x_p)}] = \pi/\sqrt{x_p x_a}$ and its shifted counterpart in (12), the periapsis-to-apoapsis angle is

$$\frac{\ell}{2\sqrt{q}} \left(\frac{\pi}{\sqrt{x_p x_a}} + \frac{\pi}{\sqrt{(x_p-2b)(x_a-2b)}} \right) = \frac{\pi}{2} \left(1 + \frac{\ell}{A} \right).$$

Doubling proves the advance and (11). The same ratio follows by differentiating (6): $\Omega_\phi = \partial_\ell H$ and $\partial_\ell(B/2) = \beta$.

For a nonconstant radial phase, a phase-space return must occur after an integer number q of radial cycles. Angular return is then exactly $q\beta \in \mathbb{Z}$, equivalent to rational β ; lowest terms give primitivity. $\square$

4 Exceptional and limiting faces

Face	Exact interpretation
Circular, $E = E_c$, $\ell > 0$	The orbit is closed for every value of the epicyclic ratio. Its actual period is $2\pi/\Omega_\phi$; the noncircular rationality test does not apply.
Central equilibrium	At $\ell = 0$, $E = E_c = -\mu/(2b)$, the particle is stationary at the smooth center.
Center crossing	At $\ell = 0$, $E_c < E < 0$, the radius repeats after T_r , but velocity orientation reverses. The full Cartesian period is $2T_r$; the one-sided limiting ratio is $1/2$.
Signed momentum	Replacing $L > 0$ by $L < 0$ reverses Ω_ϕ . The geometry and radial formulas depend only on $\ell = L $.
Kepler limit	At fixed $\ell > 0$, $b \downarrow 0$ gives $J_r \rightarrow \mu/\sqrt{-2E} - \ell$ and $\beta \rightarrow 1$. The simultaneous $b, \ell \rightarrow 0$ corner reaches the Kepler collision singularity.

These clauses are logically necessary. In particular, calling every irrational-ratio circle nonclosed, or calling T_r the full Cartesian period at $L = 0$, would be false.

5 Finite certificate and independent validation

The deterministic certificate samples $\mu, b \in \{1, 2, 3\}$, $\ell \in \{0, 1, 2, 3\}$, and $I = kB/2$ for $k \in \{1, 2, 3\}$. Its 108 orbit cells store exact quadratic-field coefficients $a + c\sqrt{d}$, supplemented by eight boundary cells. Their closure partition is 36 circular or equilibrium, 18 radial center-crossing, 14 rational resonant, and 40 irrational rosette cells.

The producer-independent checker reconstructs every algebraic field and performs direct 90-digit period and apsidal quadratures. A separate SymPy program checks 1,099 symbolic/exact identities; two fresh paths replay the evidence byte for byte. A hostile suite rejects 87/87 repaired-hash and raw parser attacks. Both JSON and YAML use strict duplicate-key-rejecting validation; YAML anchors, aliases, merges, non-string keys, and implicit timestamp coercion are excluded. Enumeration is regression evidence only: Theorem 1 and Theorem 2 are all-parameter analytic statements.

6 Natural quantization and Route-A boundary

In Cartesian coordinates,

$$V(x) = -\frac{\mu}{b + \sqrt{b^2 + |x|^2}}$$

is a real bounded function, $-\mu/(2b) \leq V < 0$. Hence $-\hbar^2\Delta/2 + V$ is self-adjoint on $H^2(\mathbb{R}^2)$ and semibounded, by the bounded-perturbation theorem. This is a natural Schrödinger quantization, but its spectrum is not analyzed here and it is not a target Hilbert–Pólya operator.

Resonant closed orbits are intrinsic, but form continuous energy and rotational families rather than an isolated primitive ledger. Therefore the frozen evaluator assigns

$$(A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION),$$

with overall ROUTE_A_REJECTED. There is no arithmetic local carrier, dynamical determinant, target analytic bridge, or Route-B input.

7 Limitations, scope, and declarations

The scope literal is NO_BAD_EULER_OR_ROOT_NUMBER. We do not assert arithmetic local data, Euler factors, root numbers, automorphy, a target divisor law, a target functional equation, a target zero match, or a Hilbert–Pólya realization. Closed resonant tori are not relabeled as isolated arithmetic primitive owners, and classical isochrone formulas are not claimed as new.

Data and code availability. All evidence, source, strict evaluation, replay and mutation programs, three paper rounds, and the self-excluding release manifest are included in the HCS-C295 package.

Ethics. No human participants, animals, personal data, or field interventions were involved.

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Contributor roles. The route-certificate author performed derivation, software construction, validation design, and manuscript preparation.

AI-use disclosure. Generative tooling assisted drafting and code scaffolding. Exact identities, independent implementations, adversarial tests, and release hashes provide the stated verification; AI output is not treated as evidence by itself.

References

- [1] M. Hénon, “L’amas isochrone I,” *Annales d’Astrophysique* **22** (1959), 126–139; ADS bibcode 1959AnAp...22..126H.
- [2] M. Hénon, “L’amas isochrone II: Le calcul des orbites,” *Annales d’Astrophysique* **22** (1959), 491–498; ADS bibcode 1959AnAp...22..491H.
- [3] P. Ramond and J. Perez, “New Methods of Isochrone Mechanics,” *Journal of Mathematical Physics* **62** (2021), 112704; DOI [10.1063/5.0056957](https://doi.org/10.1063/5.0056957), [arXiv:2104.05643](https://arxiv.org/abs/2104.05643).
- [4] J.-B. Fouvry and S. Prunet, “Linear response theory and damped modes of stellar clusters,” *Monthly Notices of the Royal Astronomical Society* **509** (2022), 2443–2456; DOI [10.1093/mnras/stab3020](https://doi.org/10.1093/mnras/stab3020).